Vishwakarma Government Engineering College, Ahmedabad
Science and Humanities Department

Semester- 3 (Regular Students) (CL,CE,EC,IT,DS,ICT)

Assignments for the Subject: Probability and Statistics(BE03000251)

(As a part of PBL – Problem Based Learning)

Academic Year: 2026-27

Problem Based Learning Assignment 1 (CO 1, CO 5)
Basic Probability - 1

1. Define Mutually exclusive events and independent events. If A and B are independent events, where $P(A)=1/4$, $P(B)=2/3$, Find $P(A \cup B)$
2. A survey determines that in a locality, 33% go to work by Bike, 42% go by Car, and 12% use both. Find the probability that a random person selected uses neither of them.
3. In a high school graduating class of 100 students, 54 studied mathematics, 69 studied history and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that (a) The students took mathematics or history. (b) The students did not take either of these subjects. (c) The students took history but not mathematics.
4. A problem is given to 5 students P, Q, R, S, T. If the probability of solving the problem individually is 1/2, 1/3, 2/3, 1/5, 1/6 respectively, then find the probability that the at least one can solve this problem.
5. A and B take turns in throwing two dice, the first to throw 9 being awarded the first prize. Show that if A has the first throw, their chance of winning are in the ratio 9:8.
6. A speaks truth in 75% cases and B in 80% cases. Find the probability that they are likely to contradict each other in stating the same fact.
7. There are two boxes A and B containing 4 white, 3 red and 3 white, 7 red balls respectively. A box is chosen at random and a ball is drawn from it. If the ball is white, find the probability that it is from box A.
8. If 3 balls are "randomly drawn" from a bowl containing 6 white and 5 black balls, what is the probability that one of the balls is white and the other two black?
9. Suppose box A contains 4 red and 5 blue coins and box B contains 6 red and 3 blue coins. A coin is chosen at random from the box A and placed in box B. Finally, a coin is chosen at random from among that now in box B. What is the probability a blue coin was transferred from box A to box B given that the coin chosen from box B is red?
10. If 3 balls are "randomly drawn" from a bowl containing 6 white and 5 black balls, find the probability that one of the balls is white and the other two are black.

Basic Probability – 2

1. State Bayes' theorem. A company selected engineers through campus interview from a university. Out of the total selection made 60%, 30% and 10% are from Electronics, Computer and Mechanical respectively. If 9%, 20% and 60% of the selected students do not join the company, what is the probability that they are from computer stream?
2. Two boxes containing candies are placed on a table. The boxes are labeled B1 and B2. Box B1 contains 7 cinnamon candies and 4 ginger candies. Box B2 contains 3 cinnamon candies and 8 pepper candies. The boxes are arranged so that the probability of selecting box B1 is 1/3 and the probability of selecting box B2 is 2/3. Suresh is blindfolded and asked to select a candy. He will win a colour TV if he selects a cinnamon candy. What is the probability that Suresh will

win the TV (that is, he will select a cinnamon candy)?

3. For a certain binary Communication channel, the probability that a transmitted 'O' is received as 'O' is 0.95 and the probability that a transmitted '1' is received as '1' is 0.90. If the probability that a 'O' is transmitted is 0
find the probability that (a) a '1' is received . (b) a '1' is transmitted given that a '1' is received .
4. At a certain university, 4% of men are over 6 feet tall and 1% of women are over 6 feet tall. The total student population is divided in the ratio 3:2 in favor of women. If a student is selected at random from among all those over six feet tall, what is the probability that the student is a woman?
5. Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Past experience shows that 5%, 4% and 2% of the notebooks produced by these companies are defective. If a notebook was found to be defective, what is the probability that the notebook was supplied by A?
6. An insurance company insured 2000 bike drivers, 4000 car drivers and 6000 truck drivers. The probability of an accident involving a bike driver, a car driver and a truck driver is 0.10, 0.03 and 0.15 respectively. One of the insured persons meets with an accident. What is the probability that he is a bike driver?
7. **Define** a term random variable and explain different types of random variable.
8. If X is random variable denoting the sum obtained in rolling a pair of fair dice, determine its probability mass function.

9. Find the distribution function for the random variable X where Probability density is given by

$$f(x) = \begin{cases} \frac{1}{2}(x+1); & for -1 \le x \le 1 \\ 0 \;; & Otherwise \end{cases}$$

10. A random variable X has the following probability distribution:

x	0	1	2	3	4	5	6	7
p(x)	0	k	2k	2k	3k	k^2	$2k^2$	$7k^2+k$

(a) Find k and hence evaluate $P(0 < X < 5)$

(b) Find $P\left(\frac{1.5<X<4.5}{X>2}\right)$ (c) Find E(X)

11. If X is a continuous random variable with the following probability density function $f(x) = \begin{cases} a(2x - x^2), & 0 < x \le 2 \\ 0, & x > 2 \end{cases}$

(i) **Find** the value of " a". (ii) **Find** P[X >1].

12. A random variable X has the following probability distribution

Values of X	0	1	2	3	4	5	6	7	8
p(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

(i) **Determine** the value of a.
(ii) **Find** P(x<3); p(X ≥ 3), p (0<X<5).
(iii) **What** is the smallest value of x for which P($X < x$) > 0.5.
(iv) **Find** E(X).

Curve Fitting

1. **Find** the best-fit values of a and b so that y =a+ bx fits the data given in the table.

x	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

2. **Apply** the method of Least squares, to fit straight line y = a+ bx to the following set of observations. Also, estimate the value of y at x=72.

X	65	66	67	67	68	69	71	73
y	67	68	64	68	72	70	69	70

3. Fit a straight line y =a+ bx for the following data :

x :	1	2	3	4	5	6	7	8	9
y :	2	6	7	8	10	11	11	10	9

Also **find** difference between the actual value of y and the value of y obtained from the fitted curve in each case when x = 5.

4. In a tensile of a metal bar, the following observations were made; where x represent the load in tones and y the elongation in thousandth of millimeters:

x	1	2	3	4	5
y	35	68	100	138	170

5. **Determine** a linear relation expressing y in terms of x by the method of least squares **Explain** the method of least squares. **Apply** the method of Least squares to fit straight line to the following set of observations

X	1	1.5	2	2.5	3	3.5	4
Y	1.1	1.3	1.6	2	2.7	3.4	4.1

6. **Find** the best-fit values of a, b and c so that y =a+ bx+cx^2 fits the data given in the table.

x	1	2	3	4	5
y	35	68	100	138	170

7. **Apply** the method of Least squares to fit the second degree parabola using the least square method to the following data. Also, **estimate** y at x=6.

x	1	2	3	4	5
y	5	12	26	60	97

8. **Determine** the constants a and b by the least-square method such that $y = ae^{bx}$ fits the following data

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

9. **Apply** the method of Least squares, to fit a function of the form $y = ax^b$

x	1	1.5	2.0	2.5	3.0	3.5	4.0
y	1.1	1.3	1.6	2.6	2.7	3.4	4.1

10. **Apply** the method of Least squares, to fit a function of the form $y = ab^x$ to the following data:

x :	0	1	2	3	4
y :	1	1.0	1.3	2.5	6.3

Problem Based Learning Assignment 2 (CO 2, CO 3)

Implementing descriptive statistics calculations in MS-Excel spreadsheet

Source : A Laboratory/Practical Manual for Probability and Statisticsopics

Topics

1. Some special Probability Distributions: Binomial distribution, Poisson distribution, Poisson approximation to the binomial distribution, Normal, Exponential and Gamma densities, Evaluation of statistical parameters for these distributions.(Practical number 3, 4, 5 from a **Laboratory/Practical Manual)**
2. Basic Statistics: Measure of central tendency: Moments, Expectation, dispersion, skewness, kurtosis, Linear Correlation, correlation coefficient, rank correlation coefficient, Regression. .(Practical number 6, 7, 8 from a **Laboratory/Practical Manual)**

Problem Based Learning Assignment 3 (CO 4)

Online video based learning detailed report

Topic

Applied Statistics: Formation of Hypothesis, Test of significance: Large sample test for single proportion, Difference of proportions, Single mean, Difference of means, and Difference of standard deviations. Test of significance for Small samples: t- Test for single mean, difference of means, t-test for correlation coefficients, F- test for ratio of variances, Chi-square test for goodness of fit.

All the students are hereby informed that, as a part of the Problem Based Learning (PBL) component of Probability and Statistics every student is required to complete an Online Video Based Learning Activity through Links provided in Suggested References: in **Practical number 10. 11, 12, 13, 14 from a** Laboratory/Practical Manual or from suitable MOOC courses available on the NPTEL, SWAYAM, or MIT Open Course Ware platforms.

The objective of this activity is to strengthen the understanding of Probability and Statistics and its applications in Engineering, Science, Technology, and everyday life through self-learning using high-quality online educational resources.

Students are required to:

- Complete the selected online course or study the prescribed video lectures (minimum 8–10 hours of learning).
- Prepare and submit a detailed report (10–12 pages) summarizing the concepts learnt, important applications, key takeaways, and personal review of the course.

Report Format

The submitted report should include the following:

1. Course Title
2. Name of the Platform (NPTEL/SWAYAM/MIT Open CourseWare)
3. Course Instructor
4. Duration of Learning
5. Objectives of the Course
6. Summary of Important Concepts Learnt
7. Engineering Applications
8. Personal Learning Outcomes
9. Overall Review of the Course

Students are advised to complete the activity sincerely, as it is intended to develop self-learning skills, conceptual understanding, and the ability to relate mathematical concepts to real-world engineering applications.